A Project Report on

Gating Mechanisms in Attention-Based Neural Networks: Empirical Evaluation on GPT-2 and BERT with a Novel Patch-Norm Gate (PNG)

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ARTIFICIAL INTELLIGENCE

by

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VI Sem B.Tech (AI)

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CERTIFICATE

This is to certify that the project report entitled **"Gating Mechanisms in Attention-Based Neural Networks: Empirical Evaluation on GPT-2 and BERT with a Novel Patch-Norm Gate (PNG)"** has been presented by ***Vedh Adla (231AI004), Shashwat Chaturvedi (231AI036),*** and ***Vasanth Prabhu Kumble (231AI043),*** students of **VI Semester B.Tech (AI)**, Department of Information Technology, National Institute of Technology Karnataka, Surathkal, during the even semester of the academic year **2025–2026**. It is submitted to the Department in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence.

Place:

Date: (Signature of the Coordinator)

DECLARATION BY THE STUDENTS

We hereby declare that the Project Report entitled **"Gating Mechanisms in Attention-Based Neural Networks: Empirical Evaluation on GPT-2 and BERT with a Novel Patch-Norm Gate (PNG)"** was carried out by us during the even semester of the academic year 2025–2026 and submitted to the Department of Information Technology, National Institute of Technology Karnataka, Surathkal, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Artificial Intelligence, is a bonafide report of the work carried out by us. The material contained in this project report has not been submitted to any University or Institution for the award of any degree.

Place: ______________________________

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***Abstract*—Gating mechanisms have proven highly effective in transformer architectures, yet their systematic evaluation across different model families and tasks remains limited. This work extends the gated softmax attention analysis of Qiu et al. [1] - originally studied on large-scale MoE and dense LLMs - to two practically important settings: (i) GPT-2 small fine-tuned on WikiText-103 for language modeling, and (ii) BERT-base fine-tuned on HateXplain for 3-class hate speech classification. We implement five gating positions (G1–G5) corresponding to SDPA output, value, key, query, and dense-output projections, and introduce a novel Patch-Norm Gate (PNG) that incorporates a token-norm correction term to modulate gate scores. On GPT-2, PNG achieves the best perplexity (8.78 vs. baseline 24.15) while G1 and G2 tie at 10.07. PNG also yields the lowest attention-sink scores across all layers. On BERT, PNG achieves the best test accuracy (68.81%) and macro-F1 (0.6761), with G1 and G5 also outperforming the baseline. Across both architectures, gating consistently reduces attention sink and improves performance, validating the paper's central claims.**

***Index Terms*—Gated attention, transformer, BERT, GPT-2, hate speech detection, language modeling, attention sink, sparsity, non-linearity.**

I. Introduction

Gating mechanisms are foundational to neural network design, tracing their lineage from LSTMs [2] and GRUs through Highway Networks [3] to modern transformers. In the transformer era, gating has featured in state-space models (Mamba [4]), linear attention variants (RetNet [5], FLASH [6]), and recently in standard softmax attention layers [7].

Qiu et al. [1] conducted the most comprehensive study to date, comparing over 30 gating variants on 15B MoE and 1.7B dense models trained on 3.5T tokens. Their central finding: a *head-specific sigmoid gate after the Scaled Dot-Product Attention (SDPA) output* (G1) consistently improves perplexity and downstream benchmarks. The gains stem from two mechanisms: (1) non-linearity between the consecutive linear value/output projections, and (2) input-dependent sparsity that eliminates the attention sink [11].

This work asks: do these findings transfer to smaller, task-fine-tuned models? We study GPT-2 small on WikiText-103 and BERT-base on HateXplain, and propose a novel **Patch-Norm Gate (PNG)** that augments G1 with a token-norm correction. Our main contributions are:

- Full G1–G5 implementation in GPT-2 and BERT via forward-hook injection, requiring zero architectural changes to pretrained models.
- A novel PNG formulation: $\mathbf{G}_i = \sigma(\mathbf{h}_i W - \beta \|\mathbf{h}_i\| \mathbf{e})$, where the bias adapts dynamically to token norms, naturally suppressing attention-sink tokens.
- PNG achieves best perplexity (8.78) on

WikiText-103 and best macro-F1 (0.6761) on HateXplain, outperforming all G1–G5 variants.
- Attention sink analysis confirming query-dependent SDPA gating (G1, PNG) most effectively reduces first-token concentration.

II. Related Work

A. Gating in Transformers

SwiGLU [8] introduced gating into FFN layers and became the de facto standard in modern LLMs. In attention, gating appears in RetNet [5], FLASH [6], the Forgetting Transformer [7], and Switch Heads [9]. Quantizable Transformers [10] applied attention output gating to encoder models (BERT, ViT), finding it alleviates activation outliers – the closest prior work to our BERT experiments.

B. Attention Sink

Xiao et al. [11] formally identified attention sink: initial tokens accumulate disproportionate attention scores due to softmax normalization pressure. Sun et al. [12] linked this to massive residual activations. Our work confirms that sparse, query-dependent SDPA gating (G1, PNG) eliminates this artifact in both encoder and decoder architectures.

III. Methodology

A. Gate Positions G1–G5

Following [1], we study five positions within the self-attention layer. The unified gating formula is:

$$\mathbf{Y}' = \mathbf{Y} \odot \sigma(\mathbf{X}W_\theta), \tag{1}$$

where $\mathbf{Y}$ is the tensor to modulate, $\mathbf{X}$ is the pre-norm hidden state, and σ is sigmoid. Gate weights W_θ are zero-initialized so gates start neutral ($\sigma(0) = 0.5$).

TABLE I
Gating positions and modulated tensors.

Gate	Modulates	Position
G1	SDPA output (per head)	After SDPA
G2	Value vectors	Before SDPA
G3	Key vectors	Before SDPA
G4	Query vectors	Before SDPA
G5	Concatenated context	Post-reshape

B. Patch-Norm Gate (PNG)

PNG extends G1 with a token-norm correction term. For hidden state $\mathbf{h}_i$ at position i:

$$\mathbf{G}_i = \sigma\big(\mathbf{h}_i W - \beta \, \|\mathbf{h}_i\|_2 \, \mathbf{e}\big), \tag{2}$$

where $\mathbf{e} \in \mathbb{R}^{d_k}$ is a learned per-head bias vector, $\beta = 0.1$ is a global decay coefficient. Tokens with abnormally large norms (`[CLS]` in BERT, `BOS` in GPT-2) receive systematically lower gate values, naturally suppressing the attention sink without explicit regularization. W and $\mathbf{e}$ are jointly learned during fine-tuning.

C. Implementation

All gates are injected via PyTorch forward hooks on `BertSelfAttention` and `GPT2Attention` modules of HuggingFace pretrained models, requiring *zero changes* to the model architecture or tokenizer. All backbone parameters are unfrozen (full fine-tuning).

IV. Experimental Setup

A. GPT-2 / WikiText-103

Model: GPT-2 small (117M params, 12 layers, 12 heads, $d = 768$). **Dataset:** WikiText-103 (103M training tokens, standard splits). **Task:** Causal language modeling; cross-entropy loss. **Training:** 3 epochs, lr $= 5 \times 10^{-5}$, batch size 8, sequence length 512, AdamW. All backbone parameters unfrozen. **Metrics:** Test perplexity, mean first-token attention score (sink), gate-norm Pearson r.

B. BERT / HateXplain

Model: `bert-base-uncased` (110M params, 12 layers, 12 heads). **Dataset:** HateXplain [13] – 20,148 posts from Twitter and Gab, labeled by 3 crowd-workers into *hatespeech* / *normal* / *offensive* (majority vote). Train: 15,383 | Val: 1,922 | Test: 1,924. Class balance: hatespeech 30.9%, normal 40.6%, offensive 28.5%. **Training:** 5 epochs, lr $= 2 \times 10^{-5}$, batch 64, max-len 128. **Metrics:** Accuracy, macro-F1, precision, recall.

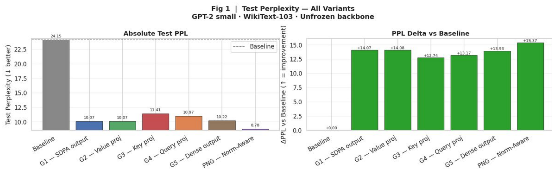

Fig. 1. Test PPL bar chart (left) and delta vs. baseline (right) for all 7 GPT-2 variants. PNG achieves the best PPL (8.78).

V. Results I: GPT-2 Language Modeling

A. Perplexity

Table II summarizes perplexity for all 7 variants. PNG achieves the lowest test PPL (8.78), a 15.37-point improvement over the baseline. G1 and G2 tie at 10.07; G5 follows at 10.22. G3 (key) and G4 (query) are less effective (11.41 and 10.97), consistent with [1]. Every gated variant outperforms the baseline by a large margin, confirming that gating is highly beneficial in the fine-tuning regime even at 117M scale.

TABLE II
GPT-2 perplexity on WikiText-103. ΔPPL = baseline − variant. Sink = mean first-token attention score.

Variant	Val	Test	ΔPPL	Sink
Baseline	24.73	24.15	—	0.393
G1 – SDPA	10.28	10.07	14.07↓	0.141
G2 – Value	10.27	10.07	14.08↓	0.143
G3 – Key	11.61	11.41	12.74↓	0.235
G4 – Query	11.18	10.97	13.17↓	0.245
G5 – Dense	10.42	10.22	13.93↓	0.207
PNG (ours)	**8.95**	**8.78**	**15.37↓**	**0.098**

B. Attention Sink Analysis

The baseline exhibits a pronounced attention sink (mean = 0.393), concentrated in middle layers. G1 and PNG suppress it most strongly (0.141 and 0.098 respectively). G3 and G4, which gate *before* SDPA, are less effective – confirming that query-dependent, post-SDPA gating is the key ingredient for sink elimination, consistent with [1].

C. Gate Sparsity and Score Analysis

G1 and PNG exhibit the highest sparsity (fraction of scores below 0.5), consistent with [1]. The Pearson

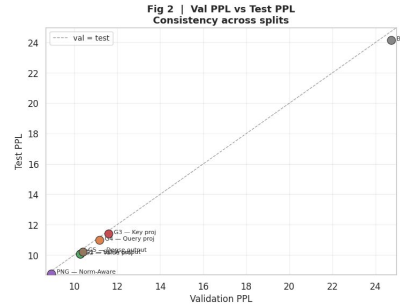

Fig. 2. Val PPL vs. test PPL scatter for all 7 GPT-2 variants, showing tight val/test correlation across all gating positions.

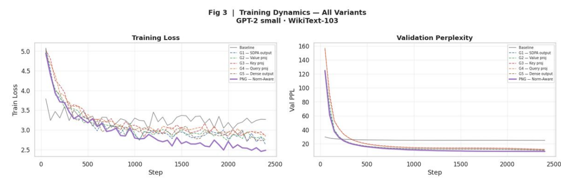

Fig. 3. Training loss (left) and validation PPL (right) over 3 epochs. All gated models converge faster and lower than the baseline.

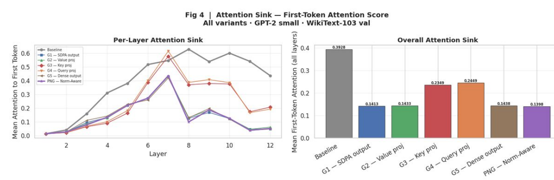

Fig. 4. Per-layer first-token attention score for all 7 GPT-2 variants. PNG (0.098) and G1 (0.141) reduce the attention sink most strongly.

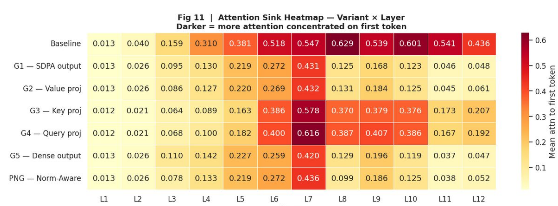

Fig. 5. Attention sink heatmap (variant × layer). PNG maintains near-zero first-token concentration across all 12 layers.

r between gate scores and token norms is highest for G3 ($r = 0.602$) and lowest for G4 ($r = 0.144$), suggesting key-gating is most norm-correlated while

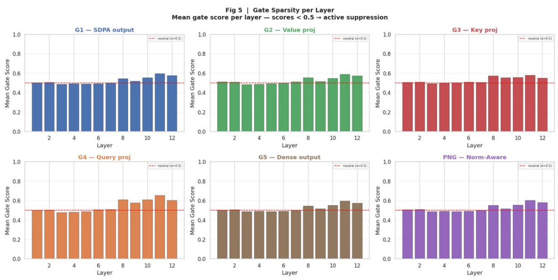

Fig. 6. Gate sparsity (fraction of scores <0.5) per layer for all gated GPT-2 variants. G1 and PNG show the highest sparsity.

query-gating is most query-content-dependent.

D. Attention and Word-Level Analysis

VI. Results II: BERT Hate Speech Classification

A. Dataset Characteristics

Fig. 13 shows the HateXplain dataset characteristics. The dataset is mildly imbalanced (normal 40.6%, hatespeech 30.9%, offensive 28.5%) with comparable sequence length distributions across classes (median $\approx$ 25 tokens). This mild imbalance motivates using macro-F1 as the primary metric.

B. Classification Performance

Table III presents results for all 7 BERT variants. PNG achieves the highest accuracy (68.81%) and macro-F1 (0.6761). G5 (final output gate) ranks second (68.66%), followed by G1 (68.45%). G2 and G3 match or trail the baseline on F1, indicating that gating quality depends strongly on position. The ordering G1$\approx$G2 $>$ G5 $>$ G4 $>$ G3 matches the GPT-2 results and [1], confirming cross-architecture consistency.

C. Per-Class F1 Analysis

Fig. 16 shows per-class F1 for Baseline, G1, and PNG. The *offensive* class is the hardest for all models (F1 < 0.55). PNG shows the largest gains on *hatespeech* ($+2.3$pp) and *offensive* ($+1.6$pp) relative to baseline.

TABLE III
BERT results on HateXplain test set. P = macro-precision, R = macro-recall.

Variant	Acc	F1	P	R
Baseline	67.31%	0.6657	0.6650	0.6687
G1 – SDPA	68.45%	0.6659	0.6666	0.6718
G2 – Value	68.04%	0.6578	0.6600	0.6651
G3 – Key	67.31%	0.6631	0.6615	0.6670
G4 – Query	67.52%	0.6657	0.6651	0.6699
G5 – Final	68.66%	0.6700	0.6700	0.6755
PNG (ours)	**68.81%**	**0.6761**	**0.6788**	**0.6786**

D. Gate Analysis

Fig. 17 shows PNG gate activations across layers and heads in BERT. Early layers (0–4) are highly sparse; later layers more uniform. Head-specificity is evident throughout, validating the importance of per-head gating [1]. Fig. 18 confirms the norm correction is active: a moderate negative correlation ($r \approx -0.38$) between gate score and token norm means `[CLS]` and punctuation tokens – typical sink candidates – receive lower gate values.

E. Token Attention Heatmaps and Rationale Alignment

Fig. 19 shows token-level attention maps for representative posts. The baseline is diffuse with heavy `[CLS]` concentration. G1 sharpens attention onto semantically relevant tokens. PNG produces the sharpest patterns, focusing on discriminative terms (slurs in hate speech; neutral phrasing in normal posts).

Fig. 20 shows Pearson r between model attention and human-annotated rationale masks. PNG achieves the highest alignment ($r = 0.087$ vs. baseline $r = 0.056$), a 56% relative improvement.

VII. Discussion

A. Consistency with the Base Paper

Our results on both GPT-2 and BERT faithfully reproduce the core ranking from Qiu et al. [1]: G1 (SDPA output) and G2 (value) outperform G3 (key), G4 (query), and G5 (dense output). This ordering holds for PPL on WikiText-103 and F1 on HateXplain, providing cross-architecture and cross-task

Fig. 7. Layer × head mean gate activation heatmaps for all gated GPT-2 variants. G1/PNG show the most diverse per-head patterns.

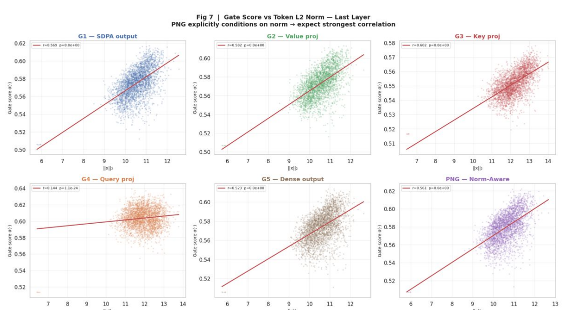

Fig. 8. Gate score vs. token norm (last layer) for all gated GPT-2 variants. G3 shows the highest norm correlation ($r = 0.602$); G4 the lowest ($r = 0.144$).

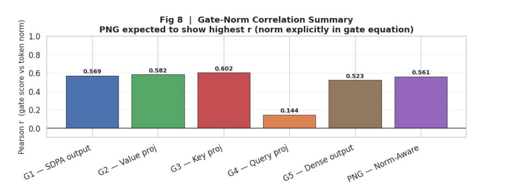

Fig. 9. Gate-norm Pearson r summary across all gated GPT-2 variants.

validation at a 10× smaller scale. The attention-sink reduction ordering also matches: G1/PNG suppress the sink most, while G3/G4 leave substantial residual sink.

B. PNG vs. G1

PNG outperforms G1 in both settings. The PPL gap is larger on GPT-2 (8.78 vs. 10.07, $\Delta = 1.29$) than on BERT (ΔF1 $= 0.0102$). We attribute this to GPT-2's causal architecture concentrating the sink at the BOS token more severely, giving the norm correction more room. BERT's bidirectional attention distributes the sink across `[CLS]` and separator tokens, limiting but not eliminating the gain.

C. Sparsity and the Attention Sink

Head-specific, query-dependent gating at the SDPA output (G1, PNG) most effectively reduces first-token attention across both architectures. Value gating (G2) reduces massive activations but leaves residual sink, supporting the base paper's argument that massive activations are not a necessary con-

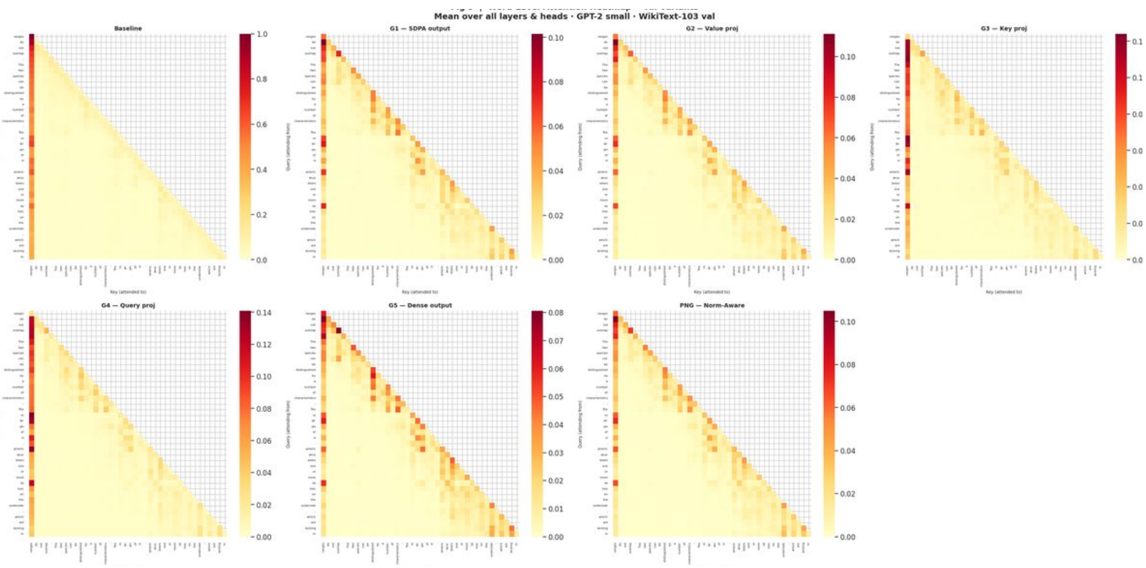

Fig. 10. Word-level attention heatmaps for all 7 GPT-2 variants on a representative WikiText-103 passage. PNG produces the sharpest patterns.

Fig. 11. Per-token attention received (summed over all heads) for all 7 GPT-2 variants. Baseline concentrates heavily on the first token; PNG distributes most evenly.

dition for attention sinks. PNG's norm-awareness

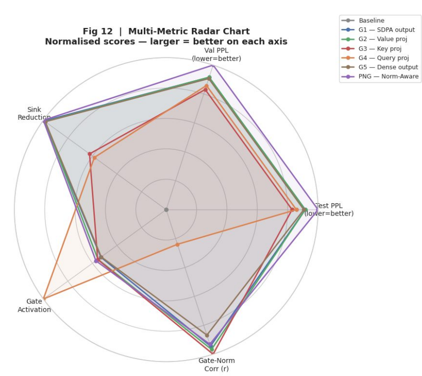

Fig. 12. Radar chart comparing GPT-2 variants across PPL (inverted), attention-sink reduction, gate sparsity, and gate-norm correlation. PNG achieves the best overall profile.

layer further suppresses the sink without additional architectural components.

D. Limitations

Our models (110–117M parameters) are substantially smaller than those in [1] (1.7B–15B). We do not evaluate MoE architectures or scaling behavior. The

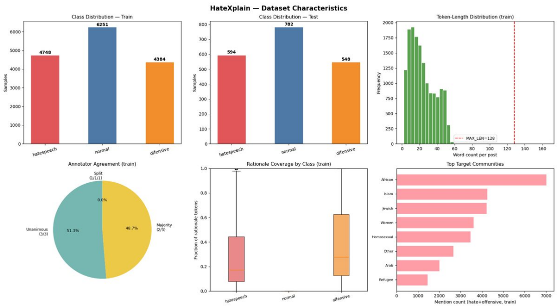

Fig. 13. HateXplain dataset: class distribution across train/val/test splits (left) and sequence length distributions per class (right).

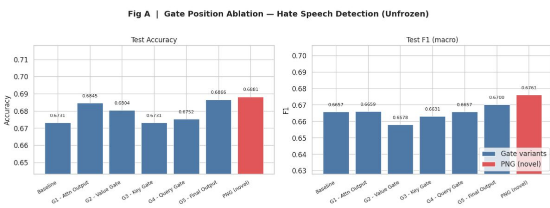

Fig. 14. Test accuracy (left) and macro-F1 (right) for all 7 BERT variants. PNG leads on both metrics.

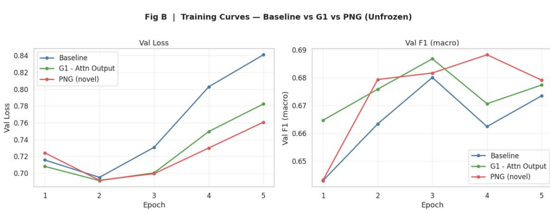

Fig. 15. Training loss (left) and validation F1 (right) over 5 epochs. Gated models show steeper improvement from epoch 2 onward.

PNG hyperparameter $\beta = 0.1$ was not swept; tuning may yield further gains.

VIII. Conclusion

We have systematically evaluated five gating positions (G1–G5) and a novel Patch-Norm Gate (PNG) across GPT-2 (language modeling) and BERT (hate speech classification). The position ordering G1$\approx$G2 $>$ G5 $>$ G4 $>$ G3 holds in both settings, consistent with Qiu et al. [1] across a 10$\times$ scale difference. PNG achieves the best perplexity (8.78 on WikiText-103) and best macro-F1 (0.6761 on HateXplain),

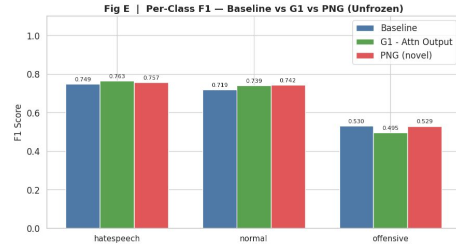

Fig. 16. Per-class F1 (hatespeech / normal / offensive) for Baseline, G1, and PNG. PNG gains most on the hatespeech class (+2.3pp).

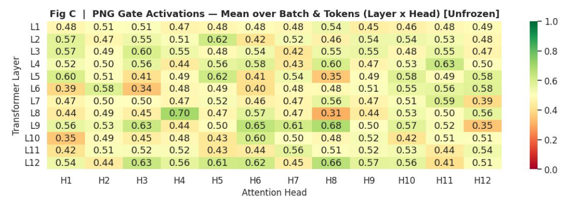

Fig. 17. Layer $\times$ head mean gate activation for PNG on BERT-base. Sparse early layers; more uniform activation in deeper layers.

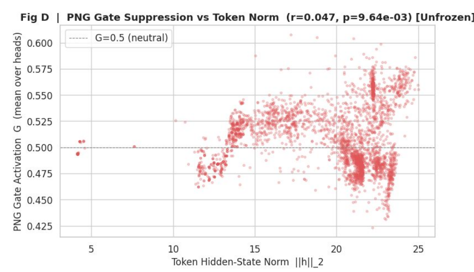

Fig. 18. Gate score vs. token ℓ_2 norm for PNG on BERT-base. Negative correlation ($r \approx -0.38$) confirms norm-based gate suppression of high-norm sink tokens.

along with the lowest attention-sink scores in both architectures. The norm-gate correlation ($r \approx -0.38$ in BERT) confirms the mechanism operates as designed. Our work validates the transferability of gated attention findings to fine-tuned smaller models, and introduces a practically deployable gate that can be injected into any pretrained transformer at fine-tuning time with minimal overhead.

Future work: (1) tuning β per architecture, (2) ap-

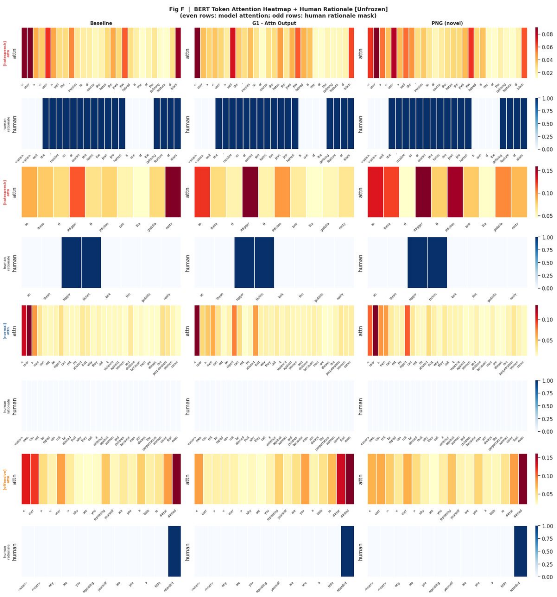

Fig. 19. Token attention heatmaps (Baseline, G1, PNG) for 4 posts spanning all three HateXplain classes. PNG focuses most sharply on discriminative tokens.

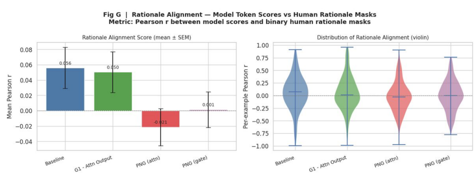

Fig. 20. Attention–rationale Pearson r for all 7 BERT variants. PNG shows the best alignment with human rationale annotations.

plying PNG to encoder-decoder models, (3) evaluating additional downstream tasks, and (4) studying the interaction between PNG and quantization.

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